

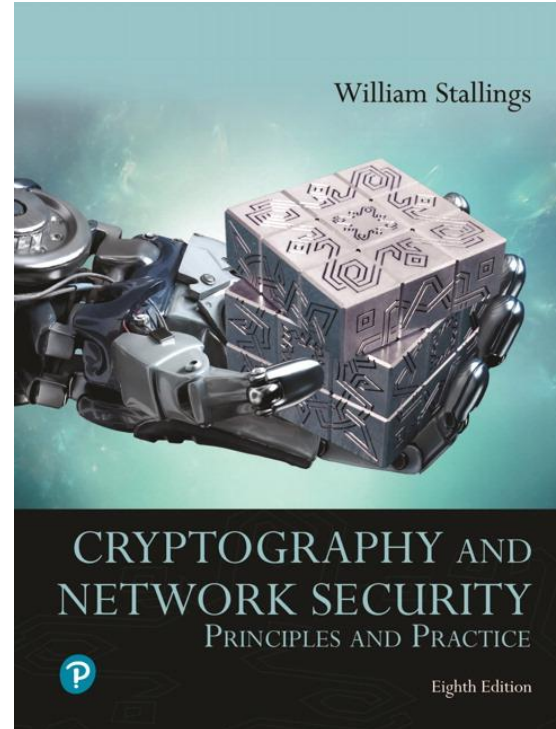
Week 10: Digital Signature

PhD. Ngoc-Tu Nguyen

tunn@uit.edu.vn

Textbooks and References

- Text books



[1] Chapter 13.14

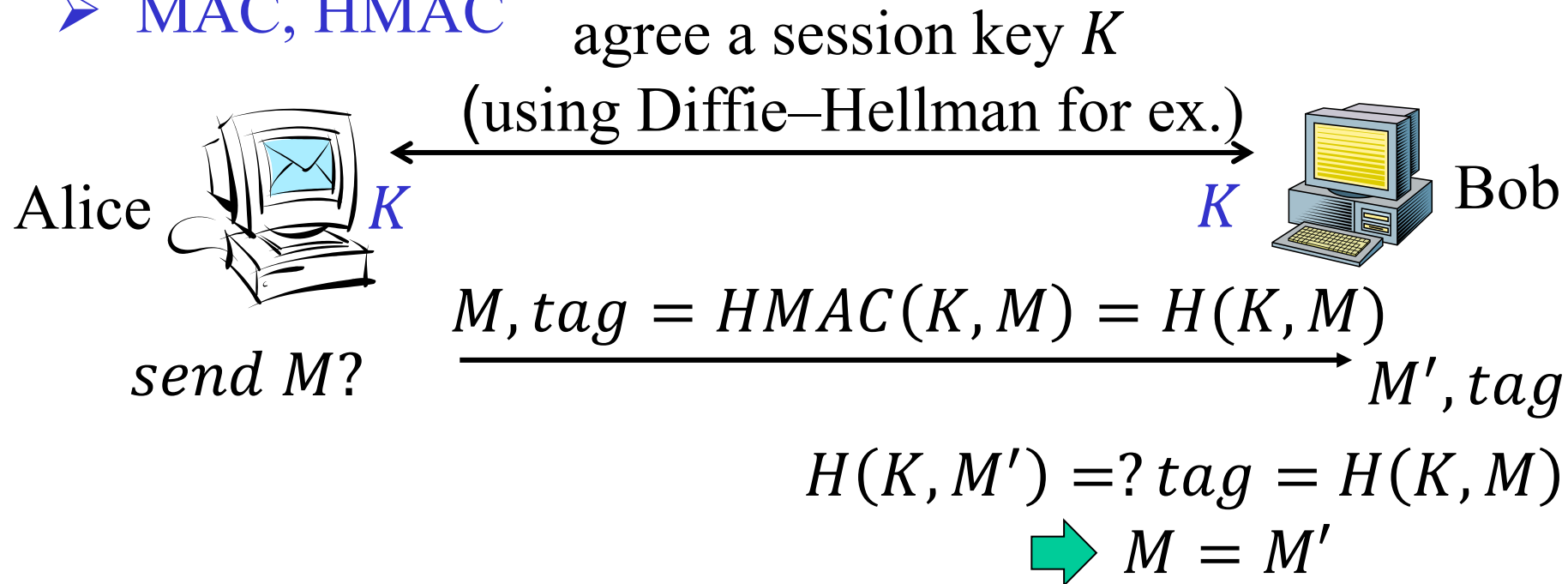
Outline

- Motivations
- Overview digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS, RSASSA-PSS
 - DSA, ECDSA
- Public key distribution (X.509 digital certificates)

Motivations

- How to ensure that the message is the original one?
 - Integrity?
- How to verify that a message comes from the claimed sender?
 - Authentication?

➤ MAC, HMAC

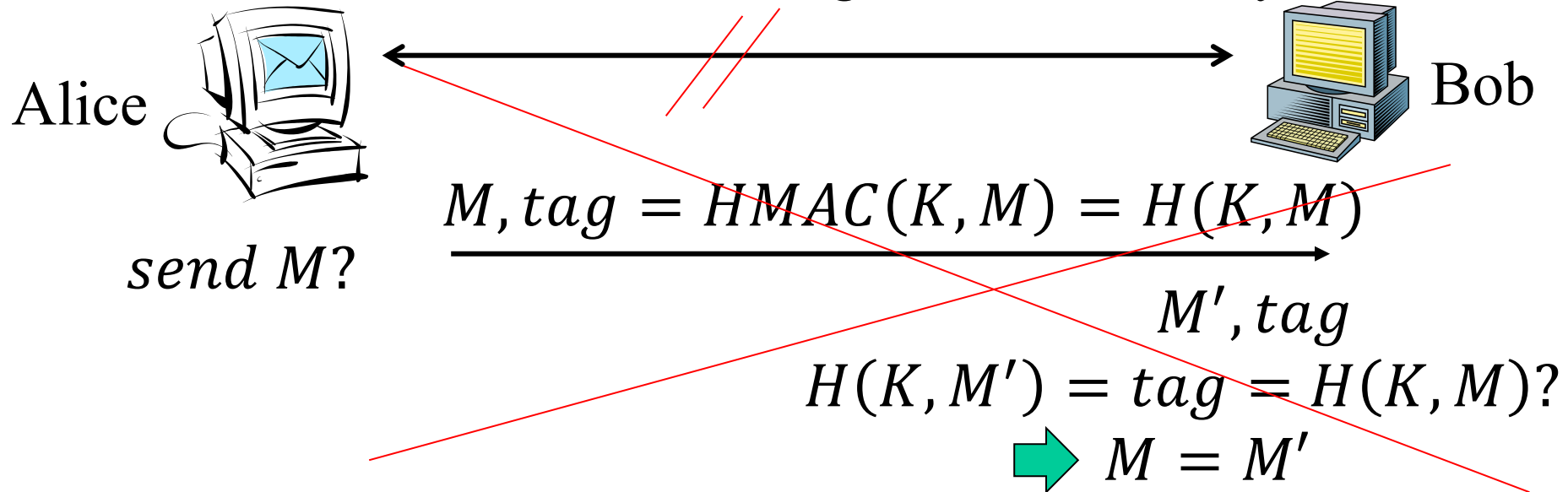


Motivations

- How to ensure that the message is the original one?
 - Integrity?
- How to verify that a message comes from the claimed sender?
 - Authentication?

➤ MAC, HMAC

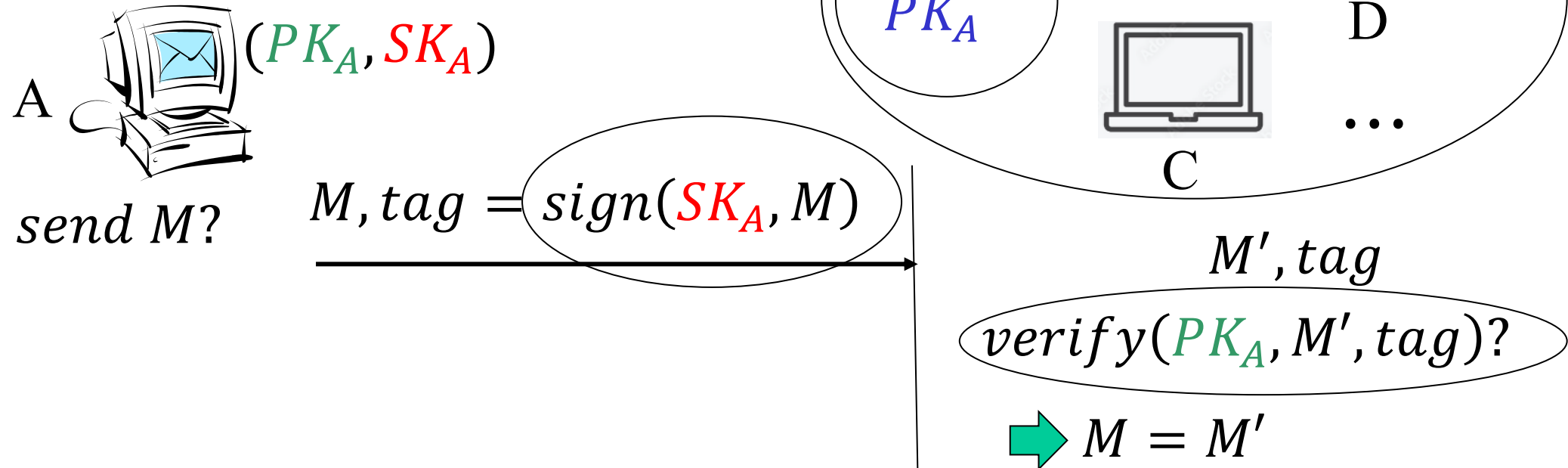
If A and B can not agree a session key K ?



Motivations

- How to ensure that the message is the original one?
 - Integrity?
- How to verify that a message comes from the claimed sender?
 - Authentication?

➤ **using digital signature**



Digital Signature Properties

- It must verify:
 - the author (signer)
 - the date and time of the signature
- It must authenticate the contents at the time of the signature(integrity);
- It must be verifiable by **third parties** (to resolve disputes);

Digital signature algorithms

1. Setup system params
2. Key generation and distribution: input λ

$$SK_A \leftarrow \text{Gen}(\lambda); PK_A \leftarrow \text{Gen}(\lambda, SK_A)$$

3. Signer signs the message

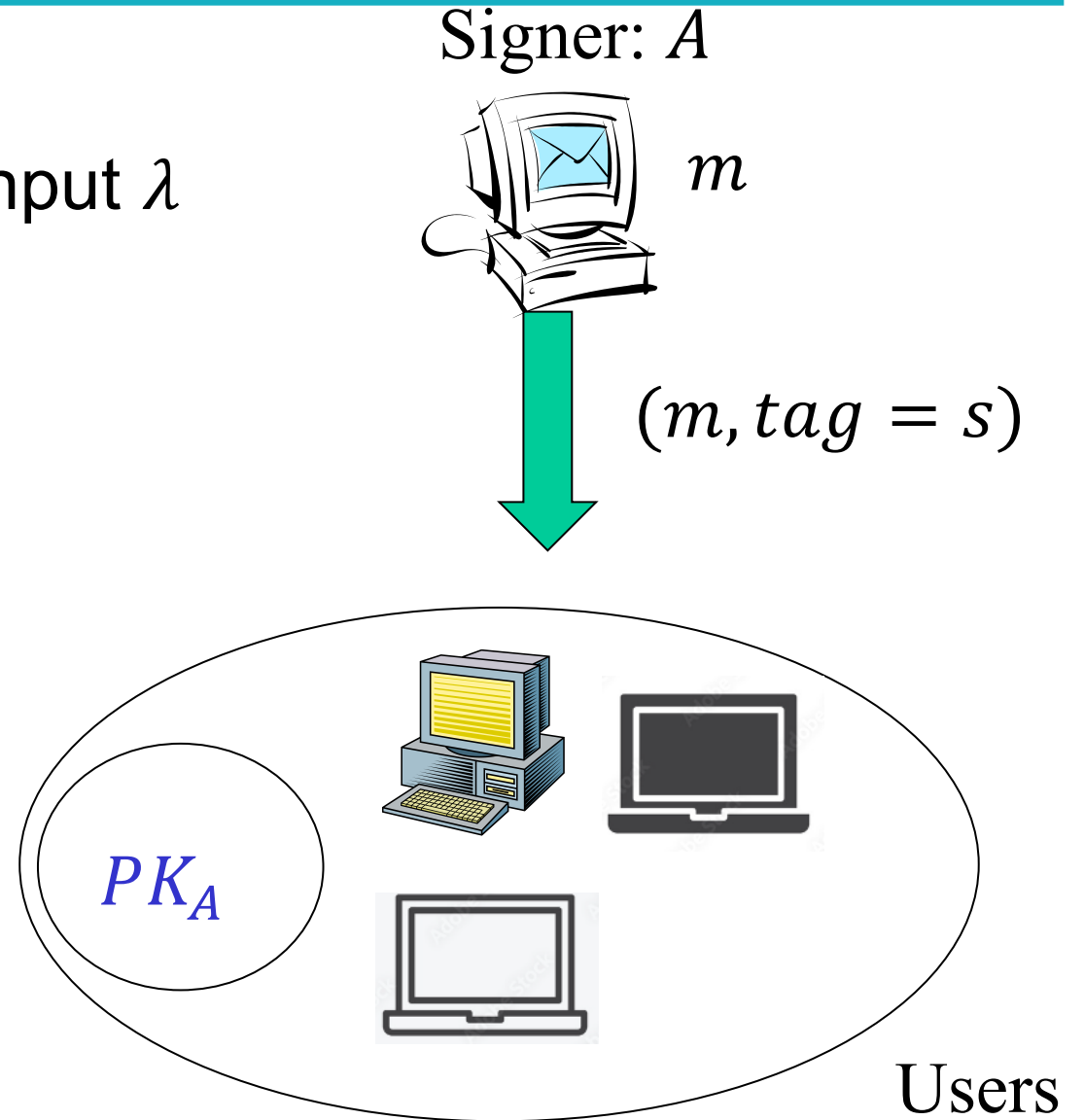
Input: m

sign: $s = \text{sign}(m, SK_A)$

4. Users verify message

Input: m, s

Verify: $\text{verify}(m, s, PK_A)$



- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS, RSASSA-PSS
 - DSA, ECDSA
- Public key distribution (X.509 digital certificates)

ElGamal Digital Signature

Parameter generation

- a key length λ ;
- λ -bit prime number p ;
- a cryptographic hash function H
$$H: \{0,1\}^* \rightarrow \{0,1\}^L \quad (L \geq \lambda)$$
- a generator g of the multiplicative group Z_p^* ;

Public parameters: **(p, g, H)**

Key generation (for signer)

- Secret key: $x \in_R Z_p^*$;
- Public key: $y = g^x \bmod p$;

Key distribution

- Verifiers have to know the signer's public key $y = g^x$;

ElGamal Digital Signature

Signing (the message m)

- Choose $k \in_R Z_p^*$
- Compute $r = g^k$
- Compute $s = (H(m) - x \cdot r) \cdot k^{-1} \bmod (p - 1)$
- Output signature (r, s)

Verifying a signature

- Input (m', r, s)
- Verify $0 < r < p, 0 < s < p - 1$
- Verify $y^r \cdot r^s \bmod p = g^{xr} \cdot g^{k \cdot (H(m) - x \cdot r) \cdot k^{-1}}$
 $= g^{H(m)} = ? g^{H(m')}$

In public key

Secret key

Outline

- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS, RSASSA-PSS
 - DSA, ECDSA
- Public key distribution (X.509 digital certificates)

Schnorr Digital Signature

Parameter generation

- prime numbers p, q , where $p = qr + 1$
- a generator g of the multiplicative group Z_p^* ;
- a cryptographic hash function H

$$H: \{0,1\}^* \rightarrow Z_q$$

Public parameters: (p, q, g, H)

Key generation (for signer)

- Secret key: $x \in_R Z_p^*$;
- Public key: $y = g^x \bmod p$;

Key distribution

- Verifiers have to know the signer's public key $y = g^x$;

Schnorr Digital Signature

Signing (the message m)

- Choose $k \in_R Z_p^*$
- Compute $r = g^k$
- Compute $e = h(r||m) \in Z_q$
- Compute $s = k - x \cdot e \bmod q = k - x \cdot h(r||m) \bmod q$
- Output signature (s, e)

Secret key In public key

Verifying a signature

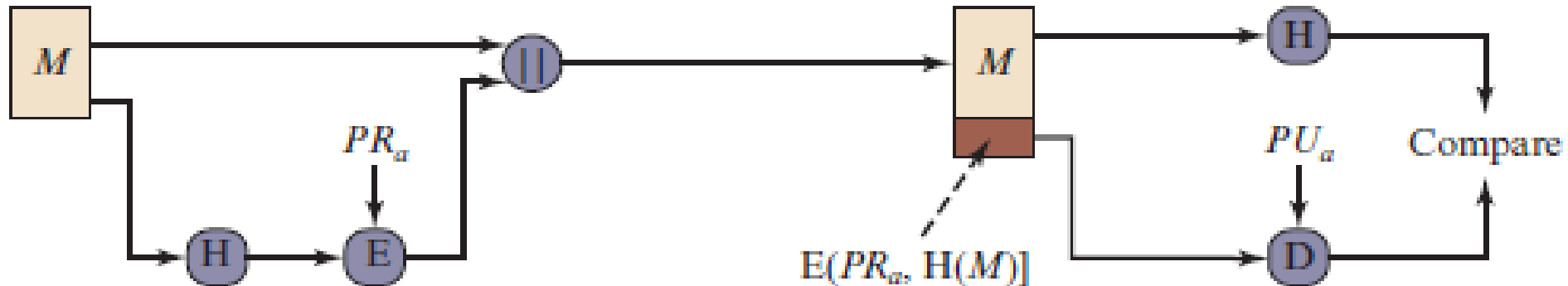
- Input (m', s, e)
- Compute $g^s \cdot y^e \bmod p = g^{k-x \cdot e} \cdot g^{x \cdot e} \bmod p$
 $= g^k = r_v$
- Verify $e = h(r||m) =? h(r_v||m')$

- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- **NIST digital signature schemes**
 - **RSASSA-PKCS, RSASSA-PSS**
 - **DSA, ECDSA**
- Public key distribution (X.509 digital certificates)

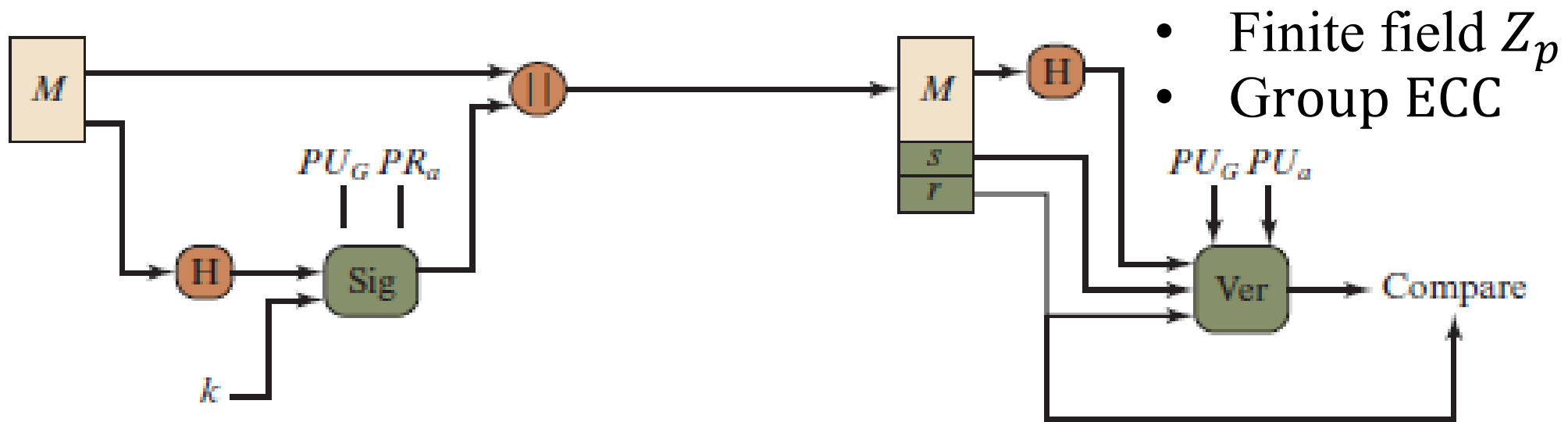
NIST Digital Signature Algorithm

- Published by NIST as Federal Information Processing Standard FIPS 186 (1994)
 - <https://nvlpubs.nist.gov/nistpubs/Legacy/FIPS/fipspub186.pdf>
 - Makes use of the Secure Hash Algorithm (SHA)
- The latest version, FIPS 186-4 (2013)
 - <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.186-4.pdf>
- Current version FIPS 186-5 (2023)
 - <https://csrc.nist.gov/publications/detail/fips/186/5/final>

Two Approaches to Digital Signatures



(a) RSA approach



(b) DSA approach

RSASSA Signatures

Raw version

- Public key is (n, e) , private key is d
- To sign message m : $s = \text{hash}(m)^d \bmod n$
 - Signing and encryption are the same mathematical operation in RSA
- To verify signature s on message m' :
$$s^e \bmod n = (\text{hash}(m)^d)^e \bmod n = \text{hash}(m) \stackrel{?}{=} \text{hash}(m')$$
 - Verification and decryption are the same mathematical operation in RSA
- Message must be padded and hashed (why?)

RSASSA Signatures

- Padding m

- Public-Key Cryptography Standards PKCS #1(v2.2):

- RSASSA-PKCS**

- <https://datatracker.ietf.org/doc/html/rfc8017>

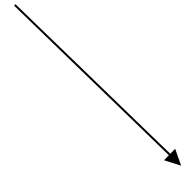
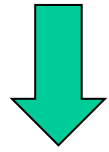
- Probabilistic Signature Scheme

- RSASSA-PSS**

- <https://datatracker.ietf.org/doc/html/rfc8017>

PKCS padding

(M, emLen)



EM = 0x00 || 0x01 || 0xff... 0xff || 0x00 || **DigestInfo value || H(M)** → RSA encrypt
signature: s



MD2: (0x)30 20 30 0c 06 08 2a 86 48 86 f7 0d 02 02 05 00 04 10 || H(M)

MD5: (0x)30 20 30 0c 06 08 2a 86 48 86 f7 0d 02 05 05 00 04 10 || H(M)

SHA-1: (0x)30 21 30 09 06 05 2b 0e 03 02 1a 05 00 04 14 || H(M)

SHA-224: (0x)30 2d 30 0d 06 09 60 86 48 01 65 03 04 02 04 05 00 04 1c || H(M)

SHA-256: (0x)30 31 30 0d 06 09 60 86 48 01 65 03 04 02 01 05 00 04 20 || H(M)

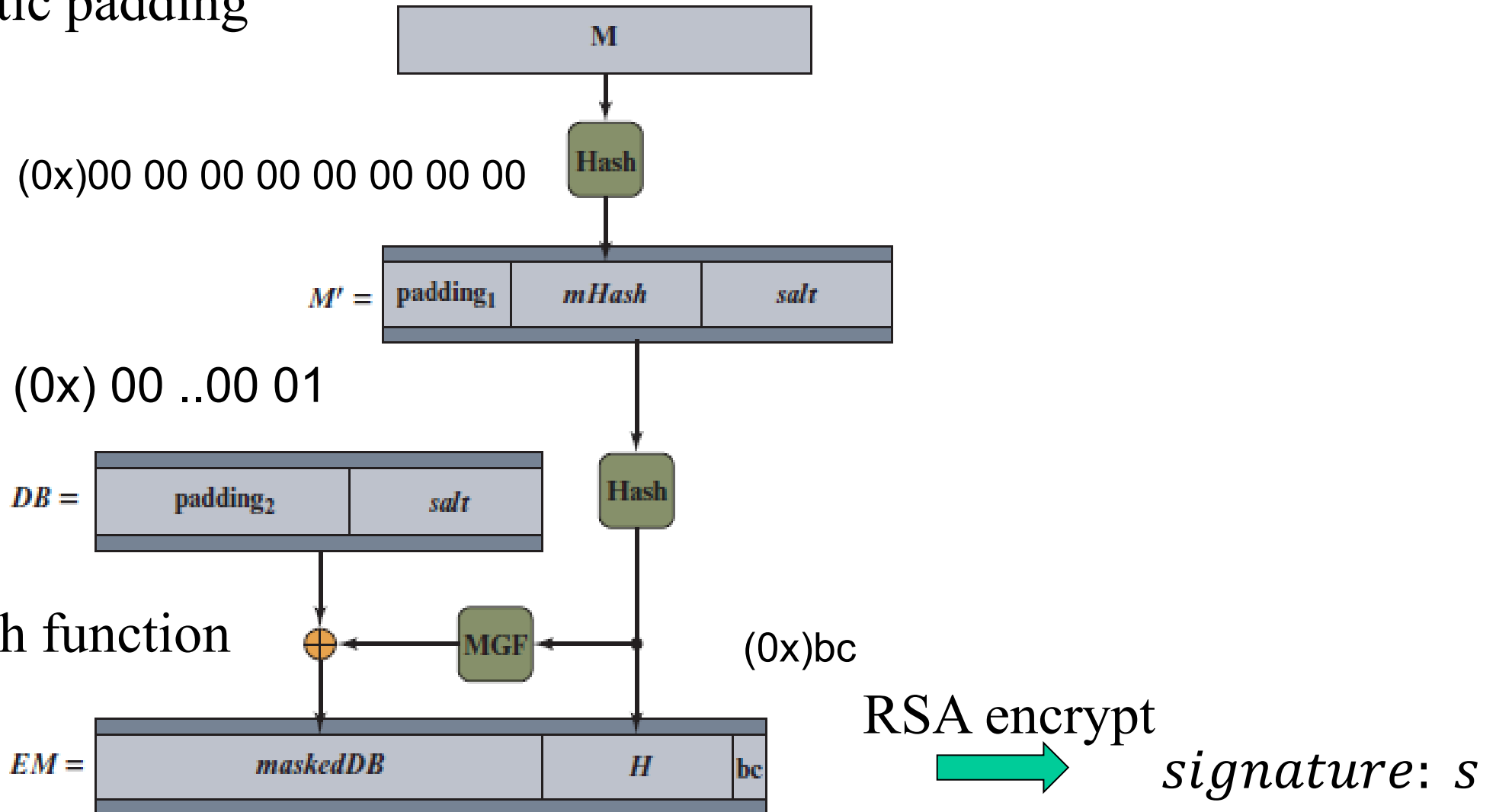
SHA-384: (0x)30 41 30 0d 06 09 60 86 48 01 65 03 04 02 02 05 00 04 30 || H(M)

SHA-512: (0x)30 51 30 0d 06 09 60 86 48 01 65 03 04 02 03 05 00 04 40 || H(M)

SHA-512/224: (0x)30 2d 30 0d 06 09 60 86 48 01 65 03 04 02 05 05 00 04 1c || H(M)

RSASSA-PSS Encoding

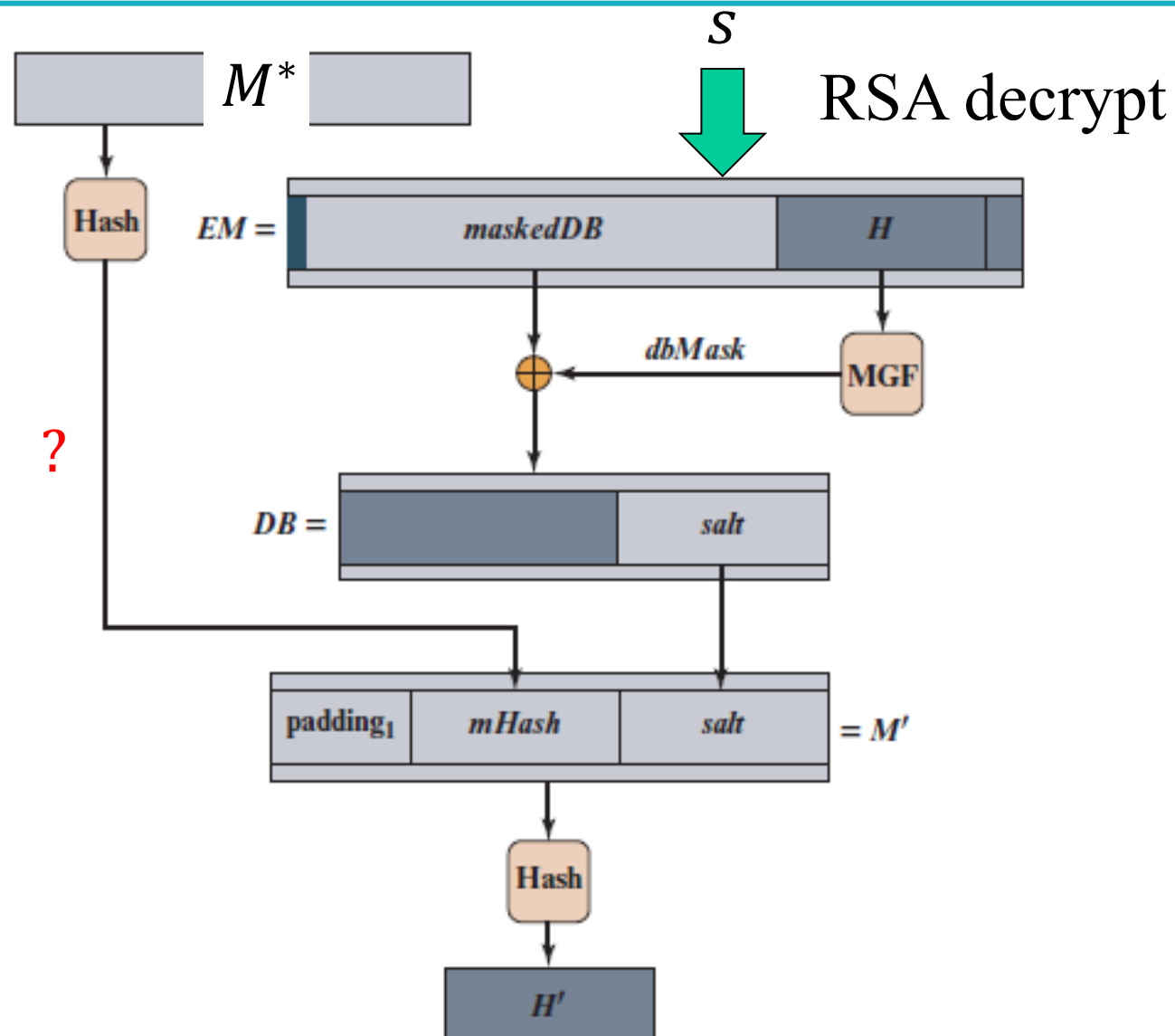
Probabilistic padding



RSASSA-PSS verifying

(M^*, s)

$H(M^*) = ? H(M) = mHash$



Outline

- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS, RSASSA-PSS
 - **DSA**, ECDSA
- Public key distribution (X.509 digital certificates)

The Digital Signature Algorithm (DSA)

➤ DSA Parameters

- p a prime modulus, where $2^{L-1} < p < 2^L$, and L is the bit length of p . Values for L are provided in Section 4.2.
- q a prime divisor of $(p - 1)$, where $2^{N-1} < q < 2^N$, and N is the bit length of q . Values for N are provided in Section 4.2.
- g a generator of a subgroup of order q in the multiplicative group of $GF(p)$, such that $1 < g < p$.
- x the private key that must remain secret; x is a randomly or pseudorandomly generated integer, such that $0 < x < q$, i.e., x is in the range $[1, q-1]$.
- y the public key, where $y = g^x \bmod p$.
- k a secret number that is unique to each message; k is a randomly or pseudorandomly generated integer, such that $0 < k < q$, i.e., k is in the range $[1, q-1]$.

The Digital Signature Algorithm (DSA)

DSA Parameters

$$2^{L-1} < p < 2^L, p - 1 : q$$

$$2^{N-1} < q < 2^N$$

$$\langle g \rangle = GF(p) = \mathbb{Z}_p$$

$$H: \{0,1\}^* \rightarrow \{0,1\}^l; l \geq \min(L, N)$$

$$L = 1024, N = 160$$

$$L = 2048, N = 224$$

$$L = 2048, N = 256$$

$$L = 3072, N = 256$$

Key generation (for signer)

Secret key: $x \in_R [1, q - 1]$

Public key: $y = g^x \bmod p \in \mathbb{Z}_p$

Key distribution

- Verifiers have to know the signer's public key $y = g^x \bmod p$;

The Digital Signature Algorithm (DSA)

Signing (the message m)

- Choose secret for each message: $k \in_R [1, q - 1]$
- Compute $r = (g^k \bmod p) \bmod q$
- Compute $s = k^{-1}(H(m) + x \cdot r) \bmod q$
- Output signature (r, s)

The Digital Signature Algorithm (DSA)

Signing (the message m)

- (m, r, s)

$$r = (g^k \bmod p) \bmod q$$

$$s = k^{-1}(H(m) + x \cdot r) \bmod q$$

Verifying a signature

- Input (m', r, s)

In public key

Secret key

- Compute:

$$w = s^{-1} \bmod q = (H(m) + x \cdot r)^{-1} \cdot k \bmod q$$

$$u_1 = H(m') \cdot w \bmod q$$

$$= H(m') (H(m) + x \cdot r)^{-1} \cdot k$$

$$u_2 = r \cdot w = r \cdot (H(m) + x \cdot r)^{-1} \cdot k \bmod q$$

- Verify $v = g^{u_1} \cdot y^{u_2} = g^{u_1} \cdot g^{x \cdot u_2} = g^{u_1 + x \cdot u_2}$
 $= g^{k(H(m') + x \cdot r)((H(m) + x \cdot r)^{-1})} = ? r$

Outline

- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS, RSASSA-PSS
 - DSA, ECDSA
- Public key distribution (X.509 digital certificates)

Elliptic Curve Digital Signature Algorithm (ECDSA)

ECDSA parameters

- Prime number: p (or $f(x)$)
- Curve coefficients: $a, b \in \mathbb{Z}_p$
- Base points: $G \in E(\mathbb{Z}_p)$
- The number $n = \text{ord}(\langle G \rangle)$
- The number $h = \frac{\text{ord}(E(\mathbb{Z}_p))}{n}$
- $H: \{0,1\}^* \rightarrow \{0,1\}^l, l = l(n)$

Key generation (for signer)

- Secret key: $d \in_R [1, n - 1]$
- Public key: $Q = d \cdot G \in E(\mathbb{Z}_p)$

Key distribution: Curve, Q

Bit length of n	Maximum Cofactor (h)	Comparable Security Strength
224 - 255	2^{14}	approximately $n/2$; at least 112 bits
256 - 383	2^{16}	approximately $n/2$; at least 128 bits
384 - 511	2^{24}	approximately $n/2$; at least 192 bits
≥ 512	2^{32}	approximately $n/2$; at least 256 bits

NIST.FIPS.186-5

Signing (the message m)

- Choose secret for each message: $k \in_R [1, n - 1]$
- Compute $R = k \cdot G = (x_1, y_1), r = x_1$
- Compute $s = k^{-1}(H(m) + d \cdot r) \bmod n$
- Output signature (r, s)

Elliptic Curve Digital Signature Algorithm (ECDSA)

Signing (the message m)

- (m, r, s)

$$r = x_1 \text{ where } kG = (x_1, y_1)$$

$$s = k^{-1}(H(m) + d \cdot r) \bmod n$$

Verifying a signature

- Input (m', r, s)
- Compute:

$$w = s^{-1} \bmod q = (H(m) + d \cdot r)^{-1} \cdot k \bmod n$$

$$u_1 = H(m') \cdot w \bmod n = H(m') (H(m) + d \cdot r)^{-1} \cdot k \bmod n$$

$$u_2 = r \cdot w = r \cdot (H(m) + d \cdot r)^{-1} \cdot k \bmod n$$

$$v_x = u_1 G + u_2 Q = (u_1 + d u_2) G$$

$$= k(H(m') + d \cdot r) ((H(m) + d \cdot r)^{-1}) G = (x'_1, y'_1)$$

- Verify $v_x = x'_1 =? r$

In public key

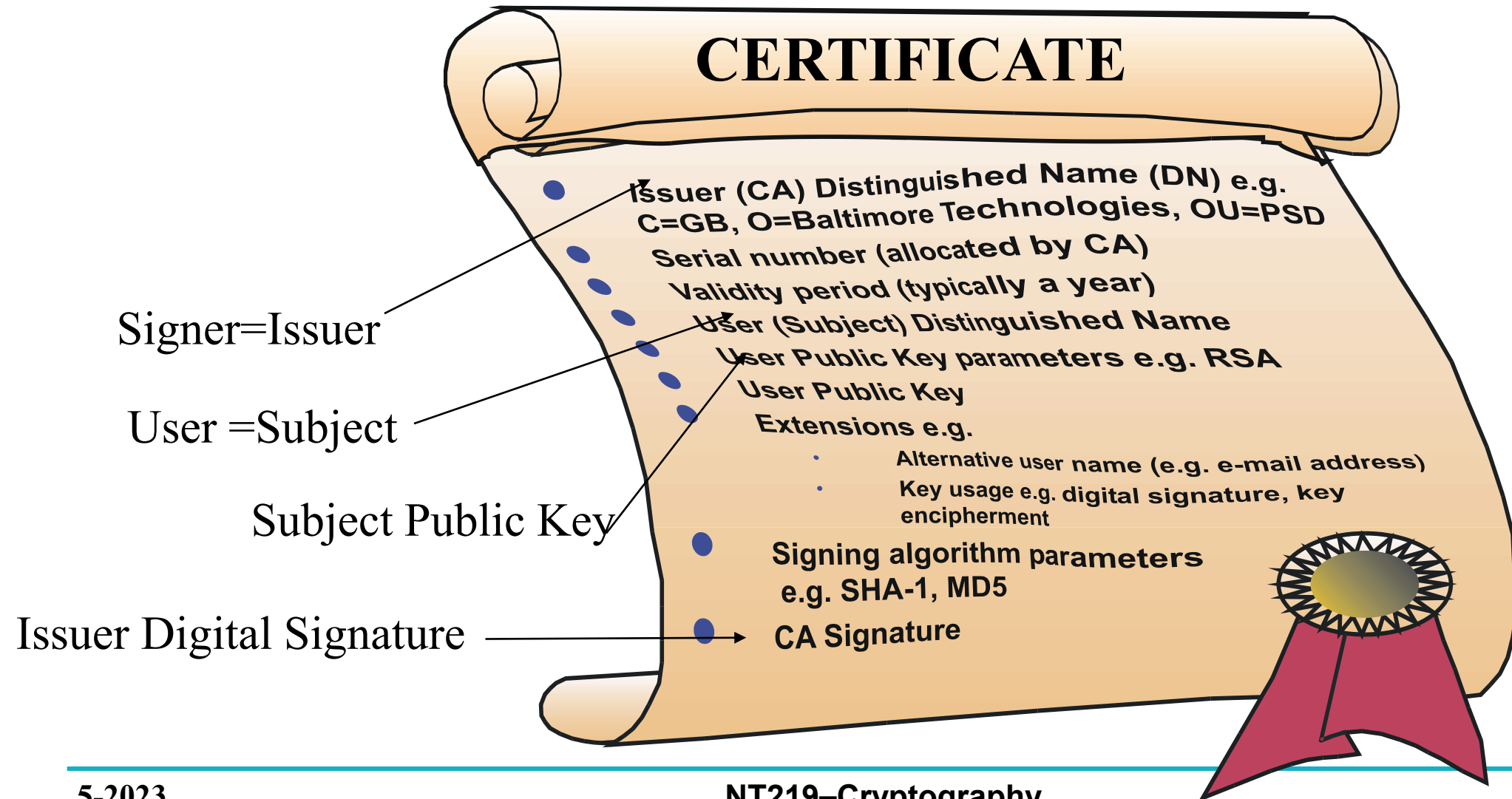
Secret key

Outline

- Motivations
- Overview of the digital signature process
- Elgamal digital signature scheme
- Schnorr digital signature scheme
- NIST digital signature schemes
 - RSASSA-PKCS
 - RSASSA-PSS
 - ECDSA
- Public key distribution (X.509 digital certificates)

X.509 digital certificate

Certificate = $(M = \{user, public\ key, \dots\}, s)$



Problems

How are Digital Certificates Issued?

Who is issuing them?

Why should I Trust the Certificate Issuer (Signers)?

How can I check if a Certificate is valid?

How can I revoke a Certificate?

Who is revoking Certificates?

X.509 digital certificate Formats

- *Version*: which version the certificate is using
- *Serial number*: a unique # assigned to the certificate within the same CA
- *Algorithm*: name of the hash function and the public-key encryption algorithm
- *Issuer*: name of the issuer
- *Validity period*: time interval when the certificate is valid
- *Subject*: name of the certificate owner
- *Public key*: subject's public-key and parameter info.
- *Extension*: other information (only available in version 3)
- *Properties*: encrypted hash value of the certificate using K_{CA}^r

Public Key Infrastructure (PKI)

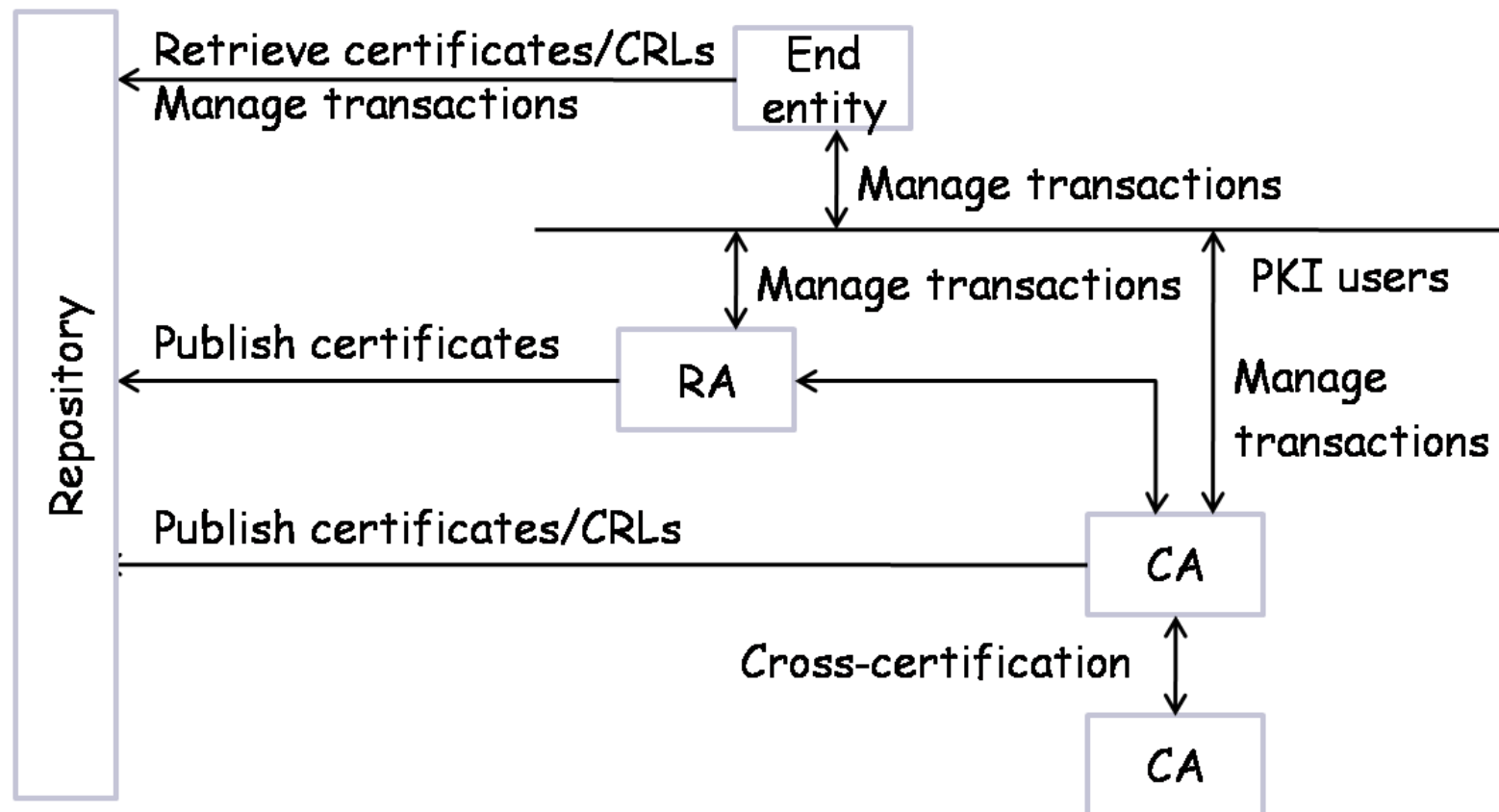
- PKI is a mechanism for using PKC
- PKI issues and manages subscribers' **public-key certificates and CA networks (signers)**:
 - Determine users' legitimacy
 - Issue public-key certificates upon users' requests
 - Extend public-key certificates' valid time upon users' requests
 - Revoke public-key certificates upon users' requests or when the corresponding private keys are compromised
 - Store and manage public-key certificates
 - Prevent digital signature singers from denying their signatures
 - Support CA networks to allow different CAs to authenticate public-key certificates issued by other CAs

X.509 PKI (PKIX)

- Recommended by IETF
- Four basic components:
 1. end entity (users, verifiers)
 2. certificate authority (CA): responsible of issuing and revoking public-key certificates;
 3. registration authority (RA): verifying identities of owners of public-key certificates
 4. Repository: responsible of storing and managing public-key certificates

<https://datatracker.ietf.org/doc/html/rfc5280>
(2008, updated 2013, 2018)

PKIX Architecture



Transaction managements:

- Registration
- Initialization
- Certificate issuing and publication
- Key recovery
- Key generation
- Certificate revocation
- Cross-certification

X.509 digital certificate

Example: check digital certificate of “facebook.com”

- **Check server certificate**

```
echo | openssl s_client -servername www.facebook.com -connect  
www.facebook.com:443 2> 1 | openssl x509 -text
```

- **Download server cert**

```
echo | openssl s_client -servername www.facebook.com -connect  
www.facebook.com:443 2>1 | openssl x509 -out facebook.cer -text
```

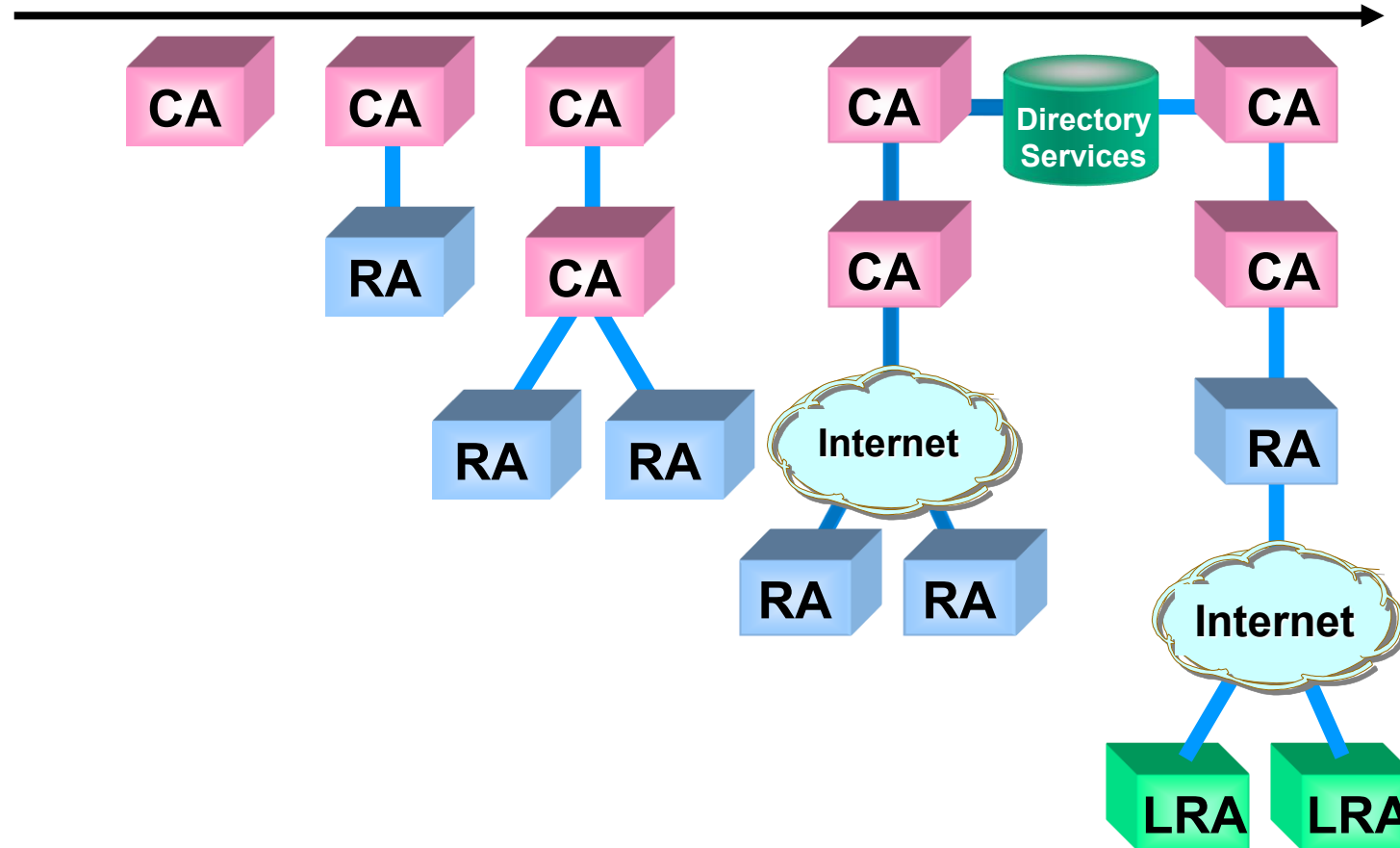
- **Check server cert file**

```
openssl x509 -in facebook.cer -inform pem -text -noout
```

https://www.openssl.org/docs/man1.1.1/man1/openssl-s_client.html

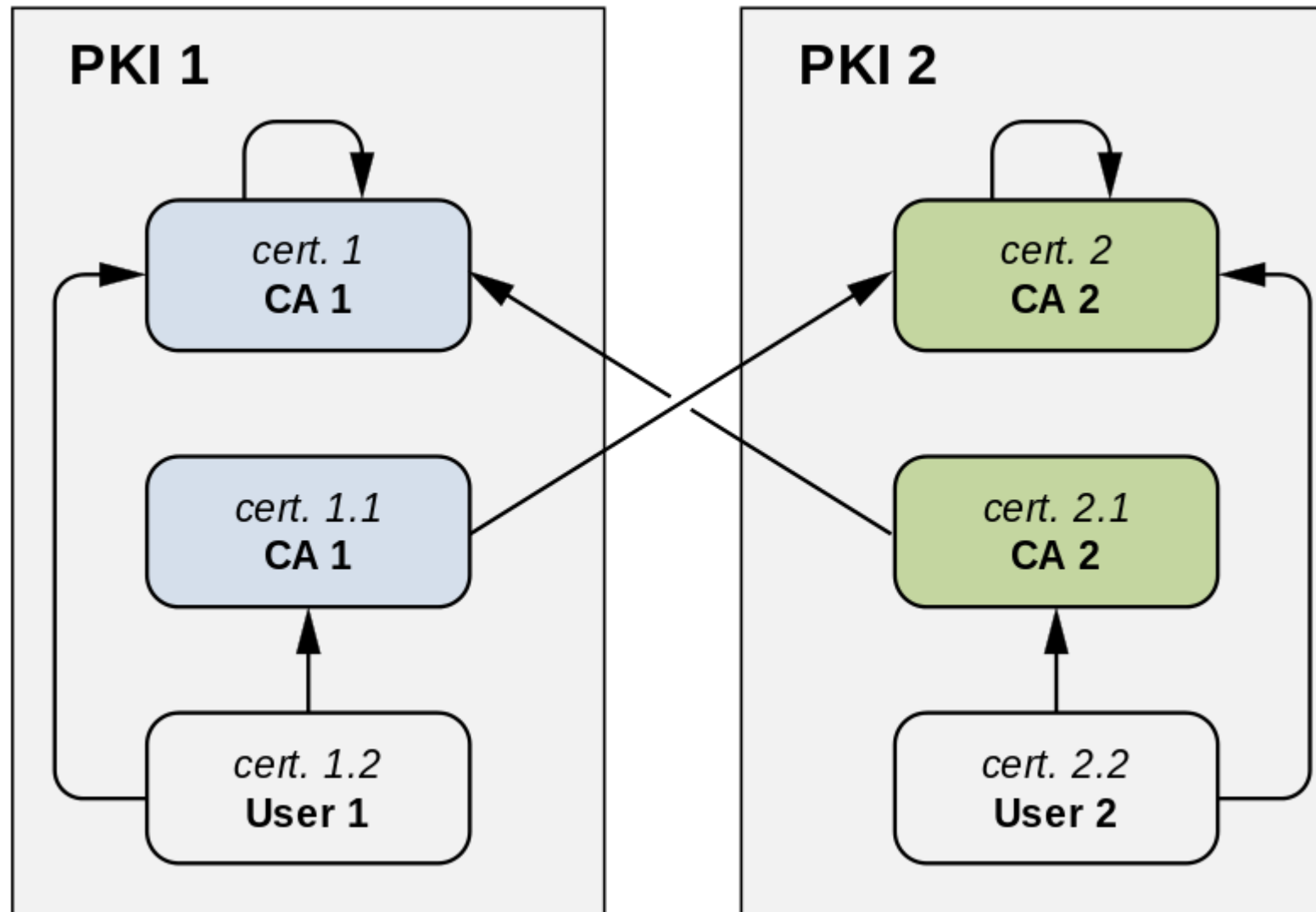
<https://www.openssl.org/docs/man1.1.1/man1/openssl-x509.html>

CA Technology Evolution



`openssl s_client -connect facebook.com:443 -showcerts`

Cross-certification between two PKIs



Attack terminology on digital signature

- **Key-only attack**

- Attackers only knows signer's public key

- **Known message attack**

- Attackers is given access to a set of messages and their signatures

■ **Generic chosen message attack**

- Attackers chooses a list of messages before attempting to break signer's signature scheme (independent of signer's public key);
- Attackers then obtains from signer valid signatures for the chosen messages;

Attack terminology on DS

- **Directed chosen message attack**

- Similar to the generic attack, except that the list of messages to be signed is chosen after attackers knows signer's public key but before any signatures are seen;

- **Adaptive chosen message attack**

- attackers may request from signer signatures of messages that depend on previously obtained message-signature pairs;

Forgery attacks

- **Total break**

- Attackers determines signer's private key;

- **Universal forgery**

- Attackers finds an efficient signing algorithm that provides an equivalent way of constructing signatures on arbitrary messages;

- **Selective forgery**

- Attackers forges a signature for a particular message chosen by signer;

- **Existential forgery**

- Attackers forges a signature for at least one message; Attackers has no control over the message;

Digital Signature Requirements

- The signature must be a bit pattern that depends on the message being signed;
- The signature must use some information unique to the sender to prevent both forgery and denial;
- It must be relatively easy to produce the digital signature;
- It must be relatively easy to recognize and verify the digital signature;
- It must be computationally infeasible to forge a digital signature, either by constructing a new message for an existing digital signature or by constructing a fraudulent digital signature for a given message;
- It must be practical to retain a copy of the digital signature in storage;

Direct Digital Signature

- **Direct Digital signature scheme:** involves only the communicating parties
 - It is assumed that the destination knows the public key of the source;
- **Confidentiality:** encrypting the entire message + signature with a shared secret key
 - It is important to perform the signature function first and then an outer confidentiality function
 - In case of dispute some third party must view the message and its signature;
- **The validity of the scheme:** depends on the security of the sender's private key
 - If a sender later wishes to deny sending a particular message, the sender can claim that the private key was lost or stolen and that someone else forged his or her signature;
 - One way to thwart or at least weaken this ploy is to require every signed message to include a timestamp and to require prompt reporting of compromised keys to a central authority